

HALOGEN🔍: Fantastic LLM Hallucinations and Where to Find Them

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Abstract

Despite their impressive ability to generate high-quality and fluent text, generative large language models (LLMs) also produce hallucinations: statements that are misaligned with established world knowledge or provided input context. However, measuring hallucination can be challenging, as having humans verify model generations on-the-fly is both expensive and time-consuming. In this work, we release **HALOGEN**🔍, a comprehensive hallucination benchmark consisting of: (1) 10,923 prompts for generative models spanning nine domains including programming, scientific attribution, and summarization, and (2) automatic high-precision verifiers for each use case that decompose LLM generations into atomic units, and verify each unit against a high-quality knowledge source. We use this framework to evaluate ~150,000 generations from 14 language models, finding that even the best-performing models are riddled with hallucinations (sometimes up to 86% of generated atomic facts depending on the domain). We further define a novel error classification for LLM hallucinations based on whether they likely stem from incorrect recollection of training data (*Type A errors*), or incorrect knowledge in training data (*Type B errors*), or are fabrication (*Type C errors*). We hope our framework provides a foundation to enable the principled study of *why generative models hallucinate*, and advances the development of trustworthy large language models.

1 Introduction

A practical challenge to deploying commercial large language models (LLMs) is their propensity to produce *hallucinated output*: facts that are not aligned with world knowledge, or with the input context provided by the user. LLM hallucinations

can cause potential downstream harms for real-world users (NIST, 2023). Yet, the reasons behind why models hallucinate are unknown. Worse, it is difficult to even measure the extent to which models hallucinate, due to the open-ended nature of model generations, and the associated time, effort, and cost of human verification.

In this work we address these challenges by (1) creating a comprehensive benchmark over diverse domains to measure hallucination behavior in language models at scale, and (2) using this diverse benchmark to investigate potential sources of language model hallucination in a range of scenarios. To estimate the degree to which LLMs hallucinate, we introduce **HALOGEN**🔍 (evaluating **H**allucinations of **G**enerative **M**odels), a large-scale evaluation suite to measure hallucination in long-form generations of LLMs (Figure 1). **HALOGEN**🔍 consists of prompts spanning nine use-cases, including tasks where a model response is expected (*response-based*) and tasks where a model is expected to abstain from answering (*refusal-based*). For each use case, we implement an *automatic verifier* that (1) decomposes a model generation into a series of meaningful atomic units specific to the use case, and (2) verifies the factuality of each atomic unit using external tools, programs, or LLM-based classifiers.

We evaluate the responses of 14 LLMs on this benchmark, spanning 150k model generations. *Our experimental results show that even the best-performing LLM responses are riddled with hallucination errors, with hallucination scores ranging from 4% to 86% depending on the task for GPT-4.* Further, we find that no single domain is highly predictive of the extent to which models will hallucinate in other domains, highlighting the need for a diverse, multi-domain benchmark such as **HALOGEN**🔍. We also find LLMs frequently hallucinate responses in scenarios where they should abstain, with even the best-performing model responding

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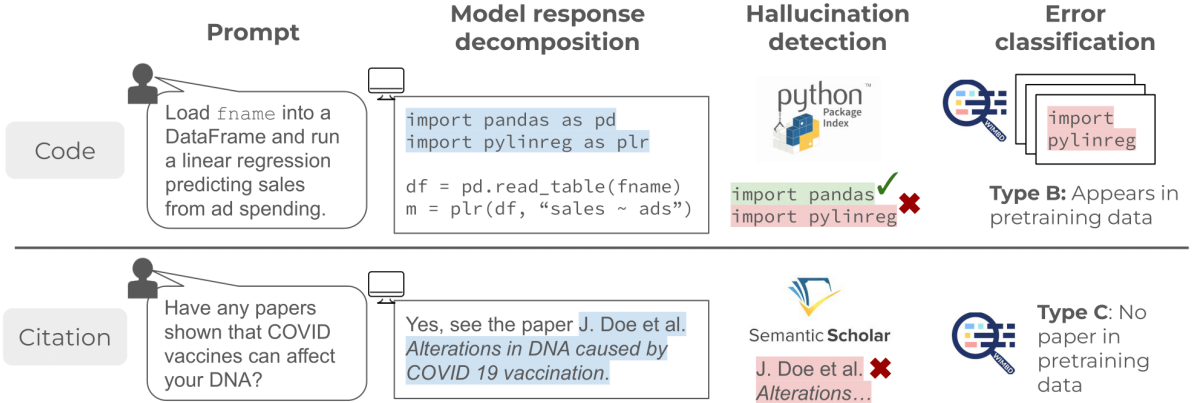


Figure 1: Hallucination evaluation for code and citation generation, two of nine evaluation settings in **HALOGEN**. Given an input prompt, we decompose each model response by identifying verifiable **atomic units**: package imports and paper citations, respectively. Then, we verify each unit to determine whether the unit is **factual** or **hallucinated**. Finally, we classify hallucinated facts into one of three categories based on relationship to training data (§1).

29% of the time, highlighting the need to improve calibration (Brahman et al., 2024).

Armed with the dataset we constructed of prompts and associated generations from several state-of-the-art language models, we trace back hallucinations to pretraining corpora. Through a series of case studies on the identified hallucinations, we isolate hallucinated atomic facts and assign error classes of the following types:

- Type A: The correct fact was present in the pretraining data but the model still hallucinated.
- Type B: An incorrect fact was in the training data, or the fact is taken out of context.
- Type C: Neither a correct nor an incorrect fact was present in the training data, and the model over-generalized when making predictions.

Our novel analysis of LLM hallucinations presents a nuanced picture. Model hallucinations do not seem to have a single isolated cause, but rather are likely to originate from a multitude of scenarios which vary across domains. For example, we find that for code-generation tasks, hallucinated software packages can often be found as-is within pretraining corpora (**Type B errors**), whereas for another task where the model hallucinates incorrect educational affiliations for US senators, the correct information is often available within the pretraining data (**Type A errors**). By providing a way to study diverse hallucination behavior in language models, and a framework for identifying the poten-

tial sources behind model hallucination, we hope to provide a systematic foundation for truthful LLMs.

2 Related Work

The tendency of LLMs to generate unfactual content, or “hallucinate”, has been well-documented in recent surveys (Zhang et al., 2023; Ji et al., 2022).

Hallucination detection Early hallucination detection work studied content-grounded tasks such as summarization (Pagnoni et al., 2021a), simplification (Devaraj et al., 2022b), and dialogue (Dziri et al., 2022). Techniques for these settings identify factual units in the model output, and compare each unit against the source text using entailment-based (Maynez et al., 2020; Kryscinski et al., 2019) or QA-based (Durmus et al., 2020) systems.

More recently, a number of works have sought to detect hallucinations occurring in open-ended generation. *Reference-based* approaches evaluate LLMs against trusted reference sources like Wikipedia or web search (Min et al., 2023; Chern et al., 2023; Mishra et al., 2024). Prior works have similarly relied on web search to identify hallucinated citations (Agrawal et al., 2023). *Reference-free* approaches instead use an LLM itself to detect hallucinations, by comparing the consistency of model responses (Manakul et al., 2023) or examining the model’s logits (Varshney et al., 2023).

Hallucination benchmarks LLM hallucination benchmarks consist of a collection of prompts designed for their potential to lead to hallucinated model output. The accuracy of the model responses

to each prompt are then evaluated, either using a more powerful LLM (Lin et al., 2021b), by examining the likelihoods assigned to correct and incorrect completions (Muhlgay et al., 2023), or by human annotators (Li et al., 2023). A number of benchmarks are also available to assess LLM factual knowledge in knowledge base completion (Mallen et al., 2022; Petroni et al., 2019) and multiple-choice (Hendrycks et al., 2020) settings.

Relative to prior benchmarks, **HALOGEN** covers a wide range of potential hallucination scenarios, including grounded generation (e.g. text summarization), open-ended generation (e.g. biographies), and bespoke use cases like scientific citation. In addition, **HALOGEN** covers both **response-based tasks**, where a model is expected to respond, and **refusal-based tasks**, where a model is expected to abstain from answering. We implement an assortment of verifiers for these use cases, ranging from entailment-based approaches for open-ended text generation to searches for Python packages and scientific references.

Factual attribution for LLMs In this work, we perform post-hoc model attribution (He et al., 2022; Gao et al., 2022) on model hallucinations. The availability of WIMBD (Elazar et al., 2023) enables us to cross-reference hallucinations with large, widely-used pretraining corpora, whereas most prior works have relied on search engines or fixed knowledge sources like Wikipedia. Model-based methods for attribution—either by prompting the model to generate citations directly (Weller et al., 2023; Khalifa et al., 2024), or via techniques like influence functions (Grosse et al., 2023)—represent an interesting future direction to better understand hallucinations observed using **HALOGEN**.

3 Building a Benchmark for Hallucinated Content

We describe the process of constructing **HALOGEN**. This benchmark consists of content-grounded tasks such as text summarization, as well as open-domain text generation tasks. For open-domain text generation, we focus on knowledge-oriented, rather than creative or subjective tasks. For instance, we do not include tasks which require a model to express a subjective opinion, engage in hyperbole, or respond creatively. We define a hallucination to be a fact in a model generation not aligned with established world knowledge or with provided context. For content-grounded tasks, we

consider hallucinations to be facts generated by a model that are not entailed by the provided context (even if they are factually correct).

It should be noted that there is no one definition of established knowledge for several facts, that truth can be pluralistic, and that data stores may contain conflicting information sources. We operationalize an ‘established’ knowledge source by specifying a singular ‘source of truth’ for each scenario, but it is possible for a practitioner to make different factuality determinations by considering different knowledge sources, or by interpreting information from the knowledge source differently.

HALOGEN includes nine tasks measuring different aspects of model factuality (Table 1). For each task, the benchmark consists of three components: (a) a set of LLM prompts X , (b) a decomposition engine D that breaks down model generations into atomic units to be verified, and (c) a hallucination detector V to automatically verify the factuality of each unit. Tasks are either **Response-Based**, where a model should provide information, or **Refusal-Based**, where a model should refuse.

3.1 Dataset

HALOGEN consists of nine tasks:

Code Packages LLMs are frequently tasked with providing coding solutions (Zhao et al., 2024b; Peng et al., 2023). Prior work has noted that generative models can hallucinate code packages, and these hallucinations can present a security vulnerability (Bar Lanyado, 2023). This study measures the extent to which models hallucinate libraries in code generation scenarios.

Prompt Construction: We obtain questions from Stack Overflow¹, based on posts in 50 different subject areas (§A).

Decomposition and Verification: We programmatically extract each imported package in the generation as an atomic unit. We then verify each package against the PyPi index.²

Summarization We quantify model hallucination in summarization, a content-grounded task where a model must synthesize salient information in provided text.

Prompt Construction: We use 1278 instances from the CNN/DailyMail dataset (Hermann et al., 2015), with instructions as shown in Table 1.

¹<https://stackoverflow.com/>

²<https://pypi.org/>